

Course Description

Module Title

English title: Data-Driven Business Modelling and Strategy

Danish title: Data-drevet forretningsmodellering og strategi

ECTS

Number of ECTS: 15

Location

Education programme: Master of Science in Economics and Business Administration (Business Data Science)

Semester: 1st

Module Coordinator

Module coordinator: Milad Abbasiharoftteh

Type and Language

Module type: Project

Language of instruction: English

Learning Objectives

KNOWLEDGE

The objective is that the student after the module possesses the necessary knowledge on:

- the representation of business processes and products in terms of predictive modelling or other data-driven processes as well as common limitations.
- how to define relevant real-world empirical problems within data driven business development including knowledge on the overall architecture and different technical elements of a data-driven project or business model.
- planning and implementation of data-driven projects in organisations, including issues within technical resources, data foundation, human resources and collaboration with external partners (customers, suppliers, knowledge organisations).

SKILLS

The objective is that the student after the module possesses the necessary skills in:

- developing and carrying out a business data science project including data collection, preprocessing, modeling, and performance assessment.
- mapping, analysing and problem-solving within data-driven projects.
- using appropriate tools (project management, business planning) for documentation.

COMPETENCES

The objective is that the student after the module possesses the necessary competences in:

- the development of data-driven projects or business opportunities from initiation to Proof of Concept and MVP stages.
- identifying challenges and developing solutions in the areas of data handling, required technical and human resources and others.
- communicating of relevant contents (e.g., business plan, project proposal) to technical, managerial, funding and general audiences.

Academic Content and Conjunction With Other Modules/Semesters

In this module students will gain knowledge of how predictive modelling and other data science techniques can be used within organisations or to develop new businesses. This includes identifying challenges when developing data-driven projects – resources, human resources (staffing), collaboration, innovation and innovation capabilities etc. Throughout the module students will explore how data-driven elements integrate with other parts of the organisation or ecosystem as well as the role of leadership of data-driven projects which requires understanding of technical and non-technical elements.

Students will work with a case (existing organisation or new business idea) and develop a business plan or project proposal covering major stages (PoC, MVP, Pilot etc.)

Based on the content above, the students will under supervision write an empirical project with an opportunity to apply a set of data science methods – a combination of techniques covered in previous courses as well as other relevant analytical approaches – to a well defined business problem. This can be in collaboration with an external organization.

After completion of the module, students are able to define an appropriate problem formulation within business data science, identify a sophisticated data collection and analysis strategy, carry out the analysis and present their results using state-of-the-art data science approaches, as well as critically self-evaluate their findings. They can select the most suitable among the wide range of methods presented in the previous modules and autonomously apply it to the specific problem under study.

Scope and Expected Performance

15 ECTS corresponds to a workload of 405 hours, distributed among the following activities:

With teacher (including guest teachers and TAs):

Lectures: 12 hours

Exercises/seminar: 12 hours

In total: 24 hours

Without teacher:

Curriculum preparation: 111 hours

Project writing with supervision: 270 hours

In total: 381 hours

Module Activities (Course Sessions etc.)

Session 1: Milad Abbasiharofteh

Introduction to the course, dynamic capabilities and innovation

- Teece, D.J. (2008). Explicating dynamic capabilities: The nature and microfoundations of (sustainable) enterprise performance. *Strategic Management Journal*, 28(13), 1319-1350.

Session 2: Milad Abbasiharofteh

Sources of innovation I: proximities and related and unrelated diversifications (Game: Eco-Innovators I)

- Balland, P.-A., Boschma, R., & Frenken, K. (2022). Proximity, innovation and networks: A concise review and some next steps. In A. Torre & D. Gallaud (Eds.), *Handbook of Proximity Relations* (pp. 70–80). Edward Elgar Publishing.

Session 3: Milad Abbasiharofteh

Sources of innovation II: collaboration and networks (Game: Eco-Innovators II) + Introduction to Project Writing — Part 1

- Abbasiharofteh, M., Kinne, J., & Krüger, M.F.C. (2023). Leveraging the digital layer: The strength of weak and strong ties in bridging geographic and cognitive distances. *Regional Studies*, 57(9), 1731-1744.

Session 4: Milad Abbasiharofteh

Dark side of innovation and eco-innovation

- Coad, A., Nightingale, P., Stilgoe, J. & Vezzani, A. (2021). Editorial: the dark side of innovation. *Industry and Innovation*, 28(1), 102-112.

Session 5: Milad Abbasiharofteh

Measuring innovation: theoretical foundations and empirical challenges

- Abbasiharofteh, M., Krüger, M., Kinne, J., Lenz, D., & Resch, B. (2023). The digital layer: alternative data for regional and innovation studies. *Spatial Economic Analysis*, 18(4), 507–529.

Session 6: Milad Abbasiharofteh

Introduction to Project Writing — Part 2: Technical, Structural and Methodological Perspectives

Slides or further information are available in Moodle.

Examination

Name of the exam: Data-Driven Business Modelling and Strategy

Type of exam:

Oral exam based on a project

Duration of the exam follows the norm catalogue for projects at AAUBS.

Assessment: 7-point grading scale Type of grading: Internal examination

Prerequisite for participation in the exam: None

Allowed aids: A computer with internet access may be used to demonstrate non-written elements of the project assignment (e.g., apps, computational notebooks, code). Otherwise, only minimal written notes (max. 10 pages) and a printed or on-screen PDF version of the project are permitted. Communication with others and the use of generative AI tools during the examination are not allowed, except when demonstrating functionality that was developed as part of the project.

Information about the re-exam:

According to the examination regulations, section 1.1 on the determination of time and place for holding exams, you as students can expect that information about the exam form, time, place, and permitted aids will be communicated no later than 2 weeks before the re-exam is held. If the re-exam is conducted digitally, this will also be communicated no later than 2 weeks before the re-exam is held.

For exam date and further information go to Moodle.